Realization cones from a finite vertex action

Human Compression project

The realization theory discussed in Section 6 of Schulte–Weiss [1] has a common linear algebraic description. Once the action on vertices is fixed, its realizations are described by positive semidefinite matrices commuting with that action. We give the argument, including the distinction between an equivariant placement, a faithful realization of faces, and a realization with a prescribed full symmetry group. The cone and its representation-theoretic interpretation are established parts of realization theory, attributed in the survey to McMullen and McMullen–Monson. They are not new theorems claimed by this project.

1 Distances and centered Gram matrices

Let X be a finite nonempty set of cardinality m, and let a finite group G act on X. Write $M = \mathbb{R}^X$ with its usual orthonormal basis $(e_x)_{x\in X}$, and let $P_g e_x = e_{gx}$. Put

$$J = I - \frac{1}{m}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad M_0 = \mathbf{1}^{\perp}.$$

A placement is a family $f = (f_x)_{x\in X}$ of points in a Euclidean space. We always replace the ambient space by the affine span of these points, and identify placements differing by an isometry that preserves the labels. Subtracting the centroid makes $\sum_x f_x = 0$. The affine span is then the linear span, denoted W.

Theorem 1. *The centered placements for which every $g \in G$ is induced by an isometry are classified, up to labeled congruence, by the cone*

$$\mathcal{C}_G = \{B \in \mathbb{R}^{X\times X} : B = B^{\mathsf{T}},\ B \succeq 0,\ B\mathbf{1} = 0,\ P_g^{\mathsf{T}} B P_g = B \text{ for every } g \in G\}.$$

The dimension of the placement is $\operatorname{rank} B$*. Its squared distance matrix is*

$$D_{xy} = B_{xx} + B_{yy} - 2B_{xy}, \qquad B = -\tfrac{1}{2}JDJ. \tag{1}$$

These formulas give mutually inverse linear maps between $\mathcal{C}_G$ and the invariant squared distance matrices.

Proof. For a placement define $T : M \to W$ by $Te_x = f_x$. Its Gram matrix is $B = T^*T$. It is positive semidefinite and symmetric, and $T\mathbf{1} = 0$ gives $B\mathbf{1} = 0$. An isometry permuting the points preserves their centroid, so after centering it is orthogonal. Its preservation of inner products is exactly the displayed invariance of B.

Conversely let $B \in \mathcal{C}_G$. The positive semidefinite square root $B^{1/2}$ commutes with every P_g, since each eigenspace of B is G-invariant. Set $W = \operatorname{im} B$ and $f_x = B^{1/2}e_x$. These vectors span W, their sum is zero, and g acts on them by the orthogonal restriction of P_g. Their Gram matrix is B.

If two spanning families f_x and f'_x have this Gram matrix, the rule $\sum_x a_x f_x \mapsto \sum_x a_x f'_x$ is well-defined: the squared norm of either sum is $a^{\mathsf{T}}Ba$. It is an onto linear isometry. Thus the matrix determines the labeled placement, including degenerate placements. Its rank is the dimension of the span.

The first formula in (1) follows by expanding $\|f_x - f_y\|^2$. Multiplying on both sides by J annihilates the two terms constant in one coordinate, and $JBJ = B$, which proves the second formula. To see that no additional condition on a proposed distance matrix has been hidden, take any symmetric hollow matrix D and set $B = -JDJ/2$. If $r_x = m^{-1}\sum_y D_{xy}$ and $s = m^{-2}\sum_{x,y} D_{xy}$, then

$$B_{xy} = -\tfrac{1}{2}(D_{xy} - r_x - r_y + s), \qquad B_{xx} = r_x - s/2.$$

Consequently $B_{xx} + B_{yy} - 2B_{xy} = D_{xy}$. If $B \succeq 0$, the construction already given realizes D. Invariance passes in both directions because J commutes with permutation matrices. □

In particular, one may specify D by one coordinate for each G-orbit on unordered pairs of distinct vertices. These are the diagonal classes used in the survey. The conditions in Theorem 1 show directly that the set of possible diagonal vectors is a closed convex cone: it is linearly isomorphic to the intersection of the positive semidefinite cone with a linear subspace. No assertion that all of its points are faithful is involved.

If f and h are two centered placements, their blend is the placement $x \mapsto (f_x, h_x)$ in $W_f \oplus W_h$, restricted to the span of these vectors. Its Gram matrix is $B_f + B_h$, and its squared distance matrix is $D_f + D_h$. The restriction to the actual span matters when the two representations contain equivalent summands. Multiplication of the matrices by $c \geq 0$ corresponds to multiplication of the placement by $\sqrt{c}$.

2 Irreducible realizations and fixed vectors

Theorem 2. *A nonzero $B \in \mathcal{C}_G$ spans an extreme ray of $\mathcal{C}_G$ if and only if the real G-module $\operatorname{im} B$ is irreducible. Every element of the cone is a sum of matrices on extreme rays.*

Proof. First suppose $W = \operatorname{im} B$ is irreducible and $B = C + D$ with $C, D \in \mathcal{C}_G$. If $v \in \ker B$, positivity gives $0 = \langle Cv, v\rangle + \langle Dv, v\rangle$, so both terms vanish. Diagonalizing a positive semidefinite matrix shows that a zero value of its quadratic form puts v in its kernel. Thus C and D vanish on $\ker B$, and symmetry makes them preserve $W = (\ker B)^{\perp}$. Every eigenspace of the symmetric operator $C|_W$ is G-invariant. Irreducibility therefore makes $C|_W$ a scalar operator. The same holds for $B|_W$, whose scalar is positive. It follows that C is a nonnegative multiple of B; the same is true of D. Hence the ray is extreme.

Conversely suppose W has a nonzero proper invariant subspace U. Its orthogonal complement is invariant, since the action is orthogonal. Let Q be orthogonal projection onto U, extended by zero on $W^{\perp}$. Then Q commutes with G. With $A = B^{1/2}$, the decomposition

$$B = AQA + A(I - Q)A$$

consists of two members of $\mathcal{C}_G$. Both are nonzero, since A is invertible on W, and their ranks are respectively $\dim U$ and $\dim W - \dim U$. Neither is proportional to B. Thus its ray is not extreme. Notice that B need not commute with an arbitrarily chosen invariant projection; the square roots on both sides of Q avoid that unjustified assumption.

For the final assertion use the placement f_x constructed in Theorem 1. Its orthogonal representation on W splits into a finite orthogonal sum of irreducible invariant subspaces: choose a minimal nonzero invariant subspace, take its orthogonal complement, and continue by induction on dimension. Projecting every f_x onto each summand gives centered equivariant placements. Their Gram matrices sum to B, and each nonzero one has that irreducible summand as its span, because the original f_x span W. The first assertion applies. □

Now assume that the action on X is transitive. Choose $x_0 \in X$ and put $H = \operatorname{Stab}_G(x_0)$, so that X is identified with the left cosets G/H. For an orthogonal representation $\rho : G \to O(V)$ write $V^H = \{v : \rho(h)v = v \text{ for every } h \in H\}$.

Proposition 3. *Equivariant placements of G/H in V are precisely*

$$f_{gH} = \rho(g)v \qquad (v \in V^H).$$

The placement is injective exactly when $\mathrm{Stab}_G(v) = H$. *Its centroid is the projection of v onto V^G. If V is nontrivial irreducible and $0 \neq v \in V^H$, the placement is centered, spans V, and gives an extreme ray of $\mathcal{C}_G$.*

Proof. The displayed expression is independent of the representative of gH exactly when v is fixed by H. Equivariance forces every placement to have this form, by taking $v = f_H$. Two cosets have the same image precisely when their quotient lies in $\mathrm{Stab}_G(v)$, which proves the injectivity assertion.

The mean of the orbit is $|G|^{-1}\sum_{g\in G}\rho(g)v$: every coset appears $|H|$ times in this sum. The averaging operator is self-adjoint and idempotent, and has image V^G, hence is the orthogonal projection onto that subspace. For a nontrivial irreducible representation $V^G = 0$, and the invariant span of the orbit of a nonzero vector is V. The last assertion follows from Theorem 2. □

This identifies the Wythoff space with a fixed-vector space. Different irreducible representations need not contribute, since V^H may be zero. Within a representation, different fixed vectors need not give the same ray. Thus an enumeration of irreducible characters alone is not a description of the realization cone. No assumption about the real, complex or quaternionic type of an irreducible representation was used. In particular, this argument does not decide Problem 23 of the survey.

3 Faces and full geometric symmetry

Let $\mathcal{P}$ be a finite abstract polytope with vertex set X. A vertex placement induces the recursive face images used in the survey: the image of a face of rank $j \geq 1$ is the set of images of its incident faces of rank $j-1$. Consider the following condition on the abstract polytope:

$$F \neq F' \text{ of rank } j \quad \Longrightarrow \quad \{E : \operatorname{rank} E = j-1,\ E < F\} \neq \{E : \operatorname{rank} E = j-1,\ E < F'\}. \quad (2)$$

It is required for every $j \geq 1$ for which there are faces to compare.

Proposition 4. *The recursive realization is faithful on every rank if and only if the vertex placement is injective and* (2) *holds. Subject to this combinatorial condition, faithfulness for $B \in \mathcal{C}_G$ is exactly $D_{xy} > 0$ for distinct vertices x, y.*

Proof. If two distinct faces have the same set of incident faces of the preceding rank, their images agree for every vertex placement. Faithfulness therefore requires (2), as well as injectivity on vertices. Conversely, suppose the images in rank $j-1$ are distinct. Equality of two sets of such images then implies equality of the two underlying sets of faces. Condition (2) makes the rank-j images distinct. Induction from rank zero proves the assertion. Finally $D_{xy} = \|f_x - f_y\|^2$, so its strict positivity is exactly vertex injectivity. □

Suppose (2) holds and the placement is faithful. Put $A = \mathrm{Aut}(\mathcal{P})$, acting on X. This action is faithful: an automorphism fixing every vertex fixes each succeeding rank by (2). The realized faces retain their ranks and incidence.

Proposition 5. *The full symmetry group of the faithful realization, in its affine span, is naturally isomorphic to*

$$A_B = \{a \in A : P_a^{\mathsf{T}} B P_a = B\}.$$

Consequently, for $G \leq A$, a faithful equivariant placement has full geometric group exactly G if and only if $B \in \mathcal{C}_G$ and no $a \in A \setminus G$ preserves B.

Proof. A symmetry of the realization permutes its faces preserving ranks and incidence. Faithfulness gives an automorphism of $\mathcal{P}$, and the symmetry preserves all squared distances. By (1) it preserves B. Conversely, if $a \in A$ preserves B, the same argument with linear relations used in Theorem 1 shows that $f_x \mapsto f_{ax}$ extends to an orthogonal isometry of the centered span. The recursive definition of face images makes it act on every face as a. Both constructions are unique because the vertices span. They preserve composition and are inverse to one another. □

The distinction in Proposition 5 is necessary for chiral realizations. A group representation and a choice of initial vertex may have more symmetry than the designated group. The proposition supplies an exact criterion for that issue for a fixed finite polytope; it asserts no existence or classification result for all polytopes.

4 Use in the source disposition

Theorem 1 proves the finite cone and diagonal-vector statements in the survey, and identifies blending with addition. Theorem 2 proves the statement about pure realizations and irreducible orthogonal representations. Proposition 3 supplies the fixed-space interpretation, with the stabilizer condition that distinguishes an injective orbit from a merely faithful representation. Propositions 4 and 5 state the further obligations needed for faithful faces and the full geometric group.

These arguments do not compute the cones for every regular toroid, classify discrete realizations in dimension three, decide the quaternionic question, or exhibit a faithful chiral realization of a regular abstract polytope. They do not replace the separately attributed geometric classifications in the survey. The present manuscript rederives common established theory; there is no new conjecture, Lean or compression-measurement credit.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Section 6 and its references to P. McMullen, *Realizations of regular polytopes*, Aequationes Math. 37 (1989), 38–56, and P. McMullen and B. R. Monson, *Realizations of regular polytopes, II*, Aequationes Math. 65 (2003), 102–112. Attribution here follows the read survey; its cited classifications are not re-established by the matrix arguments above.